

PC-05 · UV FILTERS · PERSONAL CARE

HS-BP4

Benzophenone-4

INCI	CAS	EC	FAMILY	SUPPLIED AS
Benzophenone-4	4065-45-6	223-772-2	PC-05	Per filter

01 POSITIONING

SPF economics are filter economics — broad-spectrum molecules do more of the work with less of the load.

02 HOW IT WORKS

- 01 Dual-band absorption
Extended chromophores absorb across 280–400 nm, flattening the protection curve.
- 02 Photostability
Modern broad-spectrum classes hold absorbance through solar exposure — no SPF decay, and they photostabilize fragile partners.
- 03 Load efficiency
More protection per gram means lighter textures at equal SPF.

03 APPLICATIONS

- High-SPF face and body sunscreens
- Daily-wear SPF moisturizers
- Anti-aging lines with UVA claims
- Color cosmetics with SPF

04 FORMULATION GUIDE

USE LEVEL	Per regional annexes
PH & CONDITIONS	Formulate per regional positive lists (EU Annex VI, MERCOSUR, FDA); respect ceilings per filter
— Dissolve crystalline filters hot in ester solvents (IPM/IPP class) before emulsification.	
— Combine with UVB boosters to reach target SPF at minimal total load.	

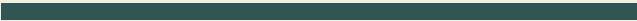
05 TYPICAL SPECIFICATION

Appearance	per filter: powder / viscous liquid
Assay	≥ 98%
Absorption range	280–400 nm
Regulatory listing	per region, per TDS

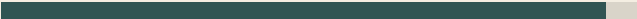
Typical parameters. The specification on the current TDS and the per-lot Certificate of Analysis govern.

06 PERFORMANCE — DIRECTIONAL

Protection breadth (UVA+UVB)
Broad-spectrum class



Photostability after 30 MED (class data)
Stabilized broad-spectrum systems



Values are representative of the ingredient class, based on published technical literature, and shown directionally for comparison. Final performance must be validated in the customer's formulation.